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Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CPLEX)

Organization	Mangalore Refinery and Petrochemicals Limited (MRPL)
Department	Mangalore Refinery and Petrochemicals Limited (MRPL)
Category	Software
Theme	Smart Automation

Background

Almost every optimization problem in India's refining, petrochemical, power, logistics, manufacturing and planning sectors ultimately depends on a handful of foreign mathematical optimization solvers such as IBM ILOG CPLEX, Gurobi and FICO Xpress. These engines sit behind refinery scheduling, production planning, supply chain optimization, blending, energy management and many AI-driven decision-support systems. While they are extremely capable, they come with high recurring license costs, restrictive licensing models and limited visibility into the underlying optimization algorithms. Indian developers can formulate optimization problems, but they cannot inspect, modify or tailor the solver internals to suit strategic national requirements. Open-source alternatives such as COIN-OR CBC, HiGHS, GLPK and SCIP exist and have made significant progress, but they still lag behind commercial solvers for several classes of large-scale mixed-integer optimization problems and have not been developed, validated or optimized specifically for Indian industrial use cases. The real challenge is not building the modeling interface; it is developing a numerically robust optimization engine that consistently finds high-quality solutions for large, sparse and highly constrained industrial problems within practical computation times.

Description

The objective is to develop a sovereign mathematical optimization solver core rather than a complete modeling environment. The solver should support Linear Programming (LP), Mixed-Integer Linear Programming (MILP) and Quadratic Programming (QP) as the initial focus, with a modular architecture that can later be extended to Mixed-Integer Quadratic Programming (MIQP), Nonlinear Programming (NLP) and Mixed-Integer Nonlinear Programming (MINLP). Core algorithms may include revised simplex and interior-point methods for continuous optimization, together with branch-and-bound, branch-and-cut, cutting planes, presolve, heuristics and advanced node selection strategies for mixed-integer problems. The solver should exploit sparse matrix techniques, efficient numerical linear algebra and multi-core parallelization, with GPU acceleration considered where it provides measurable benefits. The emphasis is on numerical stability, scalability and reliable convergence across large industrial optimization problems rather than on graphical interfaces or modelling tools. It shall not be built upon any existing open source solver library but shall be built from scratch from mathematical foundation.

The scope is to solve optimization problems arising from refinery scheduling, crude blending, process

optimization, production planning, logistics, power system dispatch, transportation and supply chain management. The benchmark is that the solver should consistently deliver optimal or near-optimal solutions for industrial-scale problems involving thousands to millions of variables and constraints, including highly degenerate models, ill-conditioned matrices and difficult mixed-integer formulations where weaker implementations exhibit excessive computation times or fail to converge.

Expected Solution

A robust optimization engine with a basic application programming interface (API) or command-line interface is sufficient; a polished graphical user interface is not required. The solver should successfully solve standard benchmark problems from recognised optimization libraries such as MIPLIB, Netlib or Mittelmann benchmark sets, with solution quality and computational performance compared against at least one established commercial or open-source solver. A clear demonstration of numerical robustness should be provided by solving challenging large-scale optimization problems involving degeneracy, weak LP relaxations or ill-conditioned constraint matrices, where simpler implementations struggle to achieve reliable convergence or acceptable solution times. The resulting solver should provide a transparent, extensible and sovereign foundation for future Indian optimization software across industrial, scientific and strategic applications.

Dataset Link

Teams to use publicly available mathematical optimization benchmark datasets such as MIPLIB, Netlib LP, Mittelmann benchmark instances, QPLIB (for quadratic programming where applicable), along with representative refinery scheduling, crude blending, production planning and supply chain optimization case studies from open literature.